

7th Week Summary (10/15/26)

- In practice, we don't know σ , so we substitute s for σ . The resulting statistic does not have a normal distribution.
- Draw a random sample of size n from a normal distribution with mean μ and standard deviation σ . The **one-sample t statistic**: $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$ has the **t distribution** with $n - 1$ degrees of freedom.

The p-value for a test of $H_0 : \mu = \mu_0$ against

$$H_a : \mu > \mu_0 \text{ is p-value} = P(T \geq t)$$

$$H_a : \mu < \mu_0 \text{ is p-value} = P(T \leq t)$$

$$H_a : \mu \neq \mu_0 \text{ is p-value} = 2P(|T| \geq |t|)$$

These p-values are exact if the population distribution is Normal; they are approximately correct for large n in other cases.

- the t distribution with $n - 1$ degrees of freedom approaches the standard normal distribution $N(0, 1)$ as n increases.
- Confidence interval for μ is $\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$ with $df = n - 1$, if
 - data is coming from normal population having unknown mean μ and unknown σ , or
 - large random sample from a non-normal population having unknown mean μ and unknown σ .

- **Comparing the means of two populations**
- **Independent samples** : A so-called two-sample problem can arise from a *randomized comparative experiment* that randomly divides subjects into two groups and exposes each group to a different treatment.

To perform inference about $\mu_1 - \mu_2$, the difference between the means of the two populations, we start from $\bar{x}_1 - \bar{x}_2$, the difference between the means of the two samples.

Two Numerical Variables – Population Standard Deviations unknown

	Independent Samples	
	$\sigma_1 = \sigma_2$	$\sigma_1 \neq \sigma_2$
Parameters of Interest:	Mean Difference, $\mu_1 - \mu_2$	Mean Difference, $\mu_1 - \mu_2$
Confidence Interval Formula:	$\bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ $df = n_1 + n_2 - 2$, where $s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}}$	$\bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ $df = \frac{(n_1-1)(n_2-1)}{(1-c)^2(n_1-1) + c^2(n_2-1)}$, where $c = \frac{\frac{s_1^2}{n_1}}{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
Name of Hypothesis Test, H_0	Two-sample T Test, $H_0 : \mu_1 = \mu_2$ or $H_0 : \mu_1 - \mu_2 = 0$	Two-sample T Test, $H_0 : \mu_1 = \mu_2$ or $H_0 : \mu_1 - \mu_2 = 0$
Test Statistic Formula:	$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$	$t' = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$
P-value:	$H_a : \mu_1 \neq \mu_2, \text{ p-value} = 2P(T \geq t)$ $H_a : \mu_1 > \mu_2, \text{ p-value} = P(T \geq t)$ $H_a : \mu_1 < \mu_2, \text{ p-value} = P(T \leq t)$	